

## TURARA RYSKULOVA, Kazakhstan

WMO: 383340

Lat: 42.5447N

Lon: 70.3539E

Elev: 808

StdP: 91.99

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.3      | -11.4 | -19.4                         | 0.7 | -12.7 | -16.7 | 1.0 | -9.4 | 10.3                  | -5.8 | 8.8 | -4.8 | 3.7                   | 80   | 0.451 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.0                   | 35.3            | 18.8 | 33.7 | 18.1 | 32.3 | 17.6 | 20.2                | 31.7 | 19.4 | 30.9 | 18.6 | 30.1 | 2.7                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.1                            | 12.6 | 25.6 | 15.0 | 11.8 | 24.7 | 14.1 | 11.1 | 24.2 | 62.0          | 32.0 | 58.9 | 30.7 | 56.2 | 30.2 | 25.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|-----|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |     | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 8.5  | 7.4 | 6.3                        | DB  | -16.3              | 37.8 | 3.6                                                | 1.5 | -18.8      | 38.9 | -20.9      | 39.8 | -22.9      | 40.7 | -25.5 | 41.8 |
| (5)               |      |     |                            | WB  | -16.7              | 22.5 | 3.5                                                | 1.4 | -19.2      | 23.5 | -21.3      | 24.3 | -23.2      | 25.1 | -25.8 | 26.1 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 12.7  | -0.4  | 0.7   | 7.7   | 12.8  | 18.2  | 22.9  | 25.5  | 24.7  | 19.8  | 13.0  | 6.4   | 0.7  | (6)  |
| (7)  |                                                                   | DBStd       | 10.33 | 5.40  | 5.93  | 5.60  | 4.76  | 3.54  | 3.04  | 2.88  | 3.20  | 4.10  | 5.19  | 5.81  | 5.68 | (7)  |
| (8)  |                                                                   | HDD10.0     | 1177  | 323   | 263   | 109   | 26    | 1     | 0     | 0     | 1     | 29    | 134   | 291   |      | (8)  |
| (9)  |                                                                   | HDD18.3     | 2737  | 580   | 493   | 329   | 172   | 47    | 3     | 0     | 1     | 31    | 174   | 358   | 547  | (9)  |
| (10) |                                                                   | CDD10.0     | 2178  | 1     | 3     | 39    | 111   | 254   | 388   | 479   | 456   | 294   | 123   | 27    | 2    | (10) |
| (11) |                                                                   | CDD18.3     | 697   | 0     | 0     | 1     | 6     | 43    | 142   | 221   | 199   | 74    | 10    | 1     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 7375  | 0     | 0     | 8     | 62    | 367   | 1437  | 2574  | 2172  | 669   | 80    | 6     | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 3147  | 0     | 0     | 0     | 7     | 82    | 563   | 1238  | 1009  | 231   | 16    | 1     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 2.6   | 2.6   | 2.8   | 3.0   | 3.0   | 2.6   | 2.4   | 2.4   | 2.8   | 2.6   | 2.3   | 2.5   | 2.7  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 676   | 77    | 81    | 81    | 86    | 68    | 30    | 21    | 9     | 13    | 56    | 71    | 81   | (15) |
| (16) |                                                                   | PrecMax     | 961   | 175   | 211   | 142   | 173   | 125   | 69    | 58    | 39    | 74    | 138   | 176   | 178  | (16) |
| (17) |                                                                   | PrecMin     | 457   | 35    | 9     | 27    | 14    | 5     | 2     | 0     | 0     | 3     | 6     | 13    |      | (17) |
| (18) |                                                                   | PrecStd     | 114   | 35    | 38    | 27    | 33    | 30    | 18    | 19    | 10    | 16    | 41    | 51    | 41   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 14.7  | 18.2  | 24.2  | 27.7  | 30.9  | 34.9  | 37.8  | 37.2  | 33.4  | 28.8  | 23.8  | 15.2 | (19) |
| (20) |                                                                   | MCWB        | 7.3   | 9.3   | 11.1  | 15.3  | 18.2  | 20.1  | 19.1  | 18.9  | 17.0  | 14.8  | 12.2  | 7.2   |      | (20) |
| (21) |                                                                   | 2%          | DB    | 10.9  | 13.6  | 21.1  | 25.2  | 28.9  | 33.1  | 35.6  | 35.3  | 31.2  | 25.4  | 19.3  | 12.6 | (21) |
| (22) |                                                                   | MCWB        | 5.2   | 7.1   | 11.0  | 15.1  | 17.8  | 18.6  | 19.0  | 18.3  | 16.1  | 13.0  | 10.3  | 6.1   |      | (22) |
| (23) |                                                                   | 5%          | DB    | 8.6   | 10.6  | 18.3  | 22.7  | 27.1  | 31.6  | 34.0  | 33.6  | 29.5  | 23.1  | 16.5  | 9.9  | (23) |
| (24) |                                                                   | MCWB        | 4.1   | 5.8   | 10.0  | 13.7  | 16.8  | 18.1  | 18.4  | 17.6  | 15.4  | 12.1  | 9.0   | 4.5   |      | (24) |
| (25) |                                                                   | 10%         | DB    | 6.4   | 8.4   | 15.9  | 20.3  | 25.3  | 30.0  | 32.5  | 32.0  | 27.3  | 20.6  | 14.0  | 7.8  | (25) |
| (26) |                                                                   | MCWB        | 2.7   | 4.5   | 9.3   | 12.8  | 15.9  | 17.6  | 18.0  | 16.8  | 14.3  | 11.2  | 7.8   | 3.4   |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 7.8   | 9.7   | 13.1  | 17.6  | 20.5  | 21.4  | 21.6  | 20.6  | 18.7  | 15.2  | 13.6  | 8.7  | (27) |
| (28) |                                                                   | MDCB        | 13.3  | 15.8  | 19.4  | 24.5  | 28.2  | 32.0  | 33.2  | 34.0  | 31.1  | 26.8  | 21.2  | 13.3  |      | (28) |
| (29) |                                                                   | 2%          | WB    | 6.0   | 7.9   | 11.8  | 16.1  | 18.8  | 20.0  | 20.2  | 19.2  | 16.7  | 13.9  | 11.4  | 6.7  | (29) |
| (30) |                                                                   | MDCB        | 10.1  | 12.3  | 18.8  | 22.7  | 27.1  | 30.6  | 32.6  | 32.8  | 29.7  | 23.9  | 17.2  | 10.9  |      | (30) |
| (31) |                                                                   | 5%          | WB    | 4.6   | 6.3   | 10.7  | 14.7  | 17.6  | 19.2  | 19.4  | 18.3  | 15.6  | 12.8  | 9.9   | 5.3  | (31) |
| (32) |                                                                   | MDCB        | 7.9   | 10.0  | 16.9  | 21.0  | 25.7  | 29.7  | 31.7  | 31.8  | 28.2  | 21.0  | 15.8  | 9.1   |      | (32) |
| (33) |                                                                   | 10%         | WB    | 3.2   | 4.7   | 9.7   | 13.5  | 16.5  | 18.3  | 18.6  | 17.5  | 14.6  | 11.7  | 8.4   | 3.9  | (33) |
| (34) |                                                                   | MDCB        | 5.9   | 8.1   | 15.4  | 19.1  | 24.2  | 28.4  | 30.9  | 30.5  | 26.3  | 19.0  | 12.9  | 7.4   |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 7.0   | 7.7   | 8.8   | 9.4   | 10.8  | 11.6  | 12.0  | 12.1  | 11.8  | 10.5  | 8.6   | 7.3   | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 9.2   | 10.2  | 12.6  | 13.0  | 12.8  | 13.1  | 13.4  | 13.5  | 13.4  | 13.1  | 11.9  | 9.6  | (36) |
| (37) |                                                                   | MCWBR       | 5.7   | 5.9   | 6.2   | 7.2   | 7.3   | 6.7   | 6.3   | 6.3   | 6.4   | 6.5   | 6.0   | 5.8   | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 8.5   | 9.3   | 10.9  | 10.9  | 11.9  | 12.0  | 12.1  | 12.5  | 12.7  | 11.5  | 10.1  | 8.8  | (38) |
| (39) |                                                                   | MCWBR       | 5.5   | 5.8   | 6.0   | 6.6   | 7.5   | 6.8   | 6.5   | 6.4   | 6.5   | 5.9   | 5.7   | 5.7   | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.338 | 0.381 | 0.454 | 0.473 | 0.451 | 0.436 | 0.423 | 0.391 | 0.375 | 0.418 | 0.396 | 0.332 | (40) |      |
| (41) |                                                                   | taud        | 2.070 | 1.983 | 1.890 | 1.924 | 2.048 | 2.122 | 2.166 | 2.245 | 2.241 | 2.022 | 2.014 | 2.089 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 760   | 787   | 773   | 796   | 828   | 840   | 847   | 864   | 850   | 736   | 678   | 726   | (42) |      |
| (43) |                                                                   | Edh at Noon | 108   | 140   | 175   | 182   | 165   | 154   | 146   | 130   | 121   | 133   | 115   | 97    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 1.86  | 2.71  | 3.59  | 4.86  | 6.33  | 7.31  | 7.37  | 6.74  | 5.29  | 3.21  | 1.87  | 1.46  | (44) |      |
| (45) |                                                                   | RadStd      | 0.24  | 0.36  | 0.32  | 0.59  | 0.49  | 0.36  | 0.37  | 0.28  | 0.25  | 0.40  | 0.28  | 0.17  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (8 neighbors) | N/A   | N/A     | N/A    | +0.66   | N/A   | N/A   | N/A         | N/A     | N/A     | +73     |

Nomenclature: See separate page